DAY 02 • STUDENT HANDOUT

Computing and Network Foundations for Threat Monitoring

Operating systems, network architecture, web servers and scripting — the knowledge base behind every alert

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats (with Common Competency ICT311203 Perform Computer Operations)
Coverage	Element 1 Required Knowledge 1.1–1.4: Windows, macOS and Linux; web server and website architecture; scripting languages; LAN, WAN and port forwarding
Duration	8 hours (1 training day)

Learning Objectives

At the end of this session, you should be able to:

1. Use basic Windows and Linux commands to check system, process and network information.
2. Describe how LANs, WANs, routers, firewalls and DMZs are arranged in an organization.
3. Explain IP addresses, ports, protocols and port forwarding.
4. Explain how a web request travels from a browser to a web server and how it is logged.
5. Read a simple script (Python / PowerShell) and explain what it does.

1. Why a Monitor Needs IT Foundations

Every alert you receive describes something that happened on an **operating system**, a **network**, or an **application**. If an alert says "powershell.exe spawned by winword.exe connecting to 203.0.113.50:443", you must understand what each part means before you can judge whether it is dangerous. The Competency Standards list these as required knowledge for checking alerts.

2. Operating Systems

	Windows	Linux	macOS
Common use	Office PCs, Active Directory servers	Web servers, cloud, security tools	Executive and design laptops
Admin account	Administrator	root (use sudo)	admin user / root
Main logs	Event Viewer: Security, System, Application	/var/log/ (auth.log, syslog)	Console app, unified log
Built-in security	Microsoft Defender, Windows Firewall	iptables/nftables, AppArmor/SELinux	XProtect, Gatekeeper

Essential Commands for Monitoring

Purpose	Windows (CMD / PowerShell)	Linux / macOS
Show IP configuration	`ipconfig /all`	`ip a` (macOS: `ifconfig`)
Show active connections	`netstat -ano`	`ss -tunap` (macOS: `netstat -an`)
List running processes	`tasklist` / `Get-Process`	`ps aux`
System information	`systeminfo`	`uname -a`, `hostnamectl`
Who is logged in	`query user`	`who`, `last`
Test connectivity	`ping`, `tracert`	`ping`, `traceroute`
DNS lookup	`nslookup example.com`	`dig example.com`

TIP — Important Windows Event IDs

- ▸ **4624** successful logon, **4625** failed logon
- ▸ **4688** new process created
- ▸ **4720** user account created, **4732** user added to a security group
- ▸ **1116 / 1117** Microsoft Defender detected / took action on malware

3. Network Infrastructure and Architecture

Concept	Description
LAN	Local Area Network — devices in one building or campus, e.g., the office network.
WAN	Wide Area Network — connects LANs over long distances, e.g., branches connected through the internet or leased lines.
Router	Connects different networks and forwards traffic between them.
Switch	Connects devices inside a LAN.
Firewall	Allows or blocks traffic based on rules (IP, port, application).
DMZ	Demilitarized zone — a separate network for public-facing servers (web, mail) so the internal LAN stays protected.
VPN	Encrypted tunnel for remote users or branches.

IP Addresses

- **Private (internal) ranges:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16 — not routable on the internet.
- **Public addresses** — reachable from the internet. Traffic from an unknown public IP to an internal server deserves attention.
- Documentation/test ranges used in our samples: 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24.

Ports and Protocols

Port	Protocol / Service	Monitoring Note
22	SSH	Remote admin on Linux; many failed logins = brute force

Port	Protocol / Service	Monitoring Note
25 / 587	SMTP	Email sending; unusual volume may mean spam bot
53	DNS	Name lookups; very long or frequent queries may indicate tunneling
80 / 443	HTTP / HTTPS	Web traffic; most malware also hides here
445	SMB	Windows file sharing; used by worms for lateral movement
3389	RDP	Remote Desktop; should never be exposed directly to the internet
3306 / 1433	MySQL / MS SQL	Databases; should not be internet-facing

Port Forwarding

Port forwarding (also called NAT rule or "publishing") tells a router/firewall to send traffic arriving on a public IP and port to a specific internal IP and port. Example: public 203.0.113.10:443 → internal 10.10.20.5:443 (web server in the DMZ). Every forwarded port is a door from the internet; an unexpected forwarding rule (e.g., to port 3389) is a red flag.

4. Web Server and Website Architecture

1. The user types a URL; **DNS** resolves the domain to an IP address.
2. The browser connects to the web server on port 443 (HTTPS).
3. The **web server** (Apache, Nginx, IIS) receives the request and passes dynamic requests to the **application** (PHP, Python, Java, .NET).
4. The application queries the **database** (MySQL, PostgreSQL, MS SQL).
5. The response goes back to the browser, and the web server writes an **access log** entry.

Reading a Web Access Log Line

```
198.51.100.23 - - [03/Nov/2026:09:15:02 +0800] "GET /login.php?user=admin' OR 1=1-- HTTP/1.1" 403 512 "-" "sqlmap/1.7"
```

Field	Value	Meaning
Client IP	198.51.100.23	Who made the request
Timestamp	03/Nov/2026 09:15:02	When
Request	GET /login.php?user=admin' OR 1=1--	Contains SQL injection pattern
Status	403	Forbidden — request was blocked
User-Agent	sqlmap/1.7	An automated attack tool — red flag

5. Scripting and Coding Basics

You do not need to be a programmer, but you must recognize what a script does because attackers use scripts (PowerShell, Python, VBScript, PHP web shells) and analysts use them to automate tasks.

Language	Common Legitimate Use	Suspicious Signs
PowerShell	Windows administration	`-EncodedCommand`, `-WindowStyle Hidden`, `DownloadString`, `IEX`
Python	Automation, security tools	Scripts in temp folders opening network connections
PHP	Websites	Unknown .php files in upload folders containing `eval(` or `base64_decode(` (web shell)
VBScript / JavaScript	Legacy automation, web pages	Attachments like invoice.vbs or .js files from email
Java	Enterprise applications	Unexpected .jar files executed from Downloads

KEY POINT

When reading any script, ask three questions: **What does it read? What does it change? Where does it connect?**

Key Terms

Term	Meaning
Port	A numbered endpoint (0–65535) identifying a network service
Port forwarding	A rule that sends traffic from a public IP/port to an internal IP/port
DMZ	Separate network segment for internet-facing servers
User-Agent	Text that identifies the software making a web request
Web shell	A malicious script uploaded to a web server to allow remote control
Event ID	Number that identifies the type of Windows log event

References

- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), units CS-ICT251101 and CS-ICT251102.
- NIST SP 800-115, Technical Guide to Information Security Testing and Assessment (Chapter 4).
- NIST SP 800-61 Rev. 2/Rev. 3, Computer Security Incident Handling Guide.
- MITRE ATT&CK® Framework — attack.mitre.org; Lockheed Martin Cyber Kill Chain®.
- Republic Act No. 10175 (Cybercrime Prevention Act of 2012) and Republic Act No. 10173 (Data Privacy Act of 2012).